

BRAIN TUMOR CLASSIFICATION REPORT

INTRODCTION

Brain tumors are the most extreme medical conditions in the medical sector, which need early and very accurate diagnosis. This projects' main aim is to create an automated system that can help the health care practitioners and medical sector to classify brain tumors using MRI scans using the deep learning techniques.

METHODOLOGY

The dataset was obtained from Kaggle which contained MRI scans which is categorized into 4 classes which is: Glioma, Meningioma, Pituitary Tumor and No Tumor. Data preprocessing included resizing images to 224x224 pixels, normalization, and data augmentation techniques such as flipping, rotation, and zooming to prevent overfitting.

Model architecture uses MobileNetV2 as the base with pre-trained ImageNet weights and the Transfer learning was applied by freezing the base model during initial training. The architecture includes global average pooling, dropout (0.2) which is for regularization, and a dense output layer with softmax activation for the 4 classes. Training was performed using the Adam optimizer with categorical crossentropy loss function for 10 epochs. The model was evaluated using accuracy, precision, recall, and F1-score metrics.

RESULTS

Models achievements:

- It achieved 86% validation accuracy
- It successfully tested on unseen images
- It deployed in multiple formats for various platforms
- It applied transfer learning, dropout, and data augmentation

In addition ,

- Precision : 0.78 macro average
- Recall : 0.72 macro average
- F1-Score : 0.73 macro average

Screenshots

```
Building SIMPLE but POWERFUL Model...
Training frozen model... (3-5 minutes)
Epoch 1/10
...
72/72 ----- 144s 2s/step - accuracy: 0.6024 - loss: 0.9560 - val_accuracy: 0.7639 - val_loss: 0.6153
Epoch 2/10
72/72 ----- 123s 2s/step - accuracy: 0.7929 - loss: 0.5410 - val_accuracy: 0.8073 - val_loss: 0.5067
Epoch 3/10
72/72 ----- 120s 2s/step - accuracy: 0.8195 - loss: 0.4782 - val_accuracy: 0.8490 - val_loss: 0.4391
Epoch 4/10
72/72 ----- 120s 2s/step - accuracy: 0.8435 - loss: 0.4224 - val_accuracy: 0.8385 - val_loss: 0.4177
Epoch 5/10
72/72 ----- 119s 2s/step - accuracy: 0.8535 - loss: 0.3808 - val_accuracy: 0.8490 - val_loss: 0.3940
Epoch 6/10
72/72 ----- 146s 2s/step - accuracy: 0.8618 - loss: 0.3489 - val_accuracy: 0.8472 - val_loss: 0.3832
Epoch 7/10
72/72 ----- 119s 2s/step - accuracy: 0.8740 - loss: 0.3379 - val_accuracy: 0.8628 - val_loss: 0.3681
Epoch 8/10
72/72 ----- 120s 2s/step - accuracy: 0.8806 - loss: 0.3122 - val_accuracy: 0.8628 - val_loss: 0.3615
Epoch 9/10
72/72 ----- 119s 2s/step - accuracy: 0.8854 - loss: 0.3012 - val_accuracy: 0.8594 - val_loss: 0.3584
Epoch 10/10
72/72 ----- 141s 2s/step - accuracy: 0.8919 - loss: 0.2810 - val_accuracy: 0.8507 - val_loss: 0.3636
```

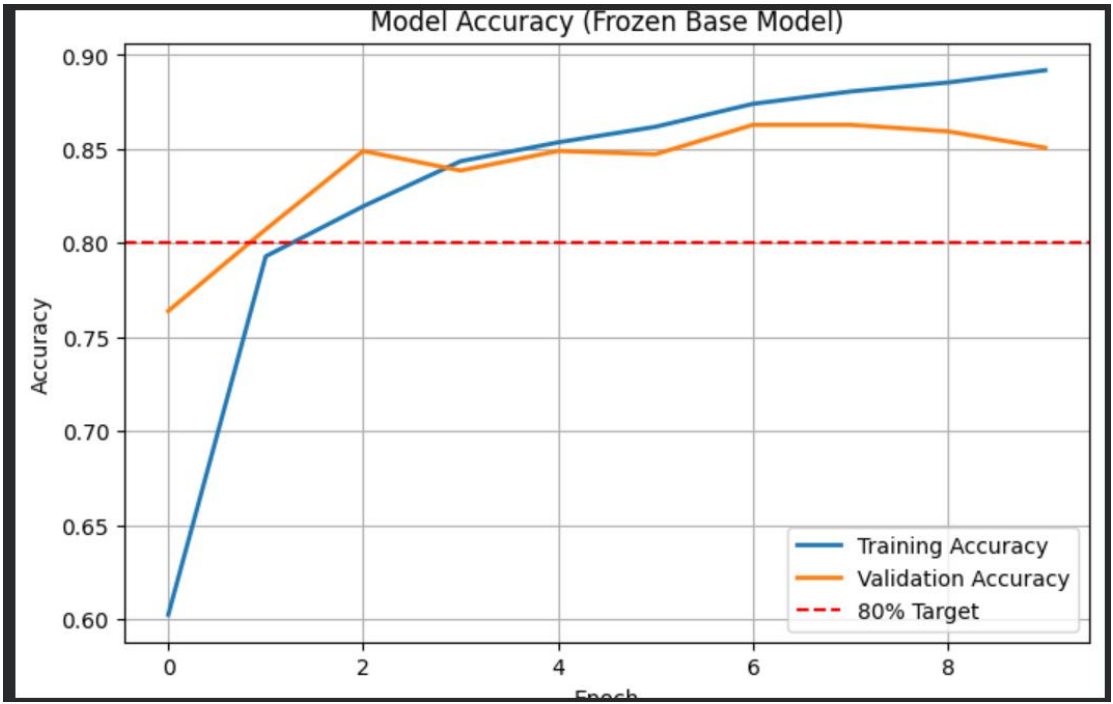
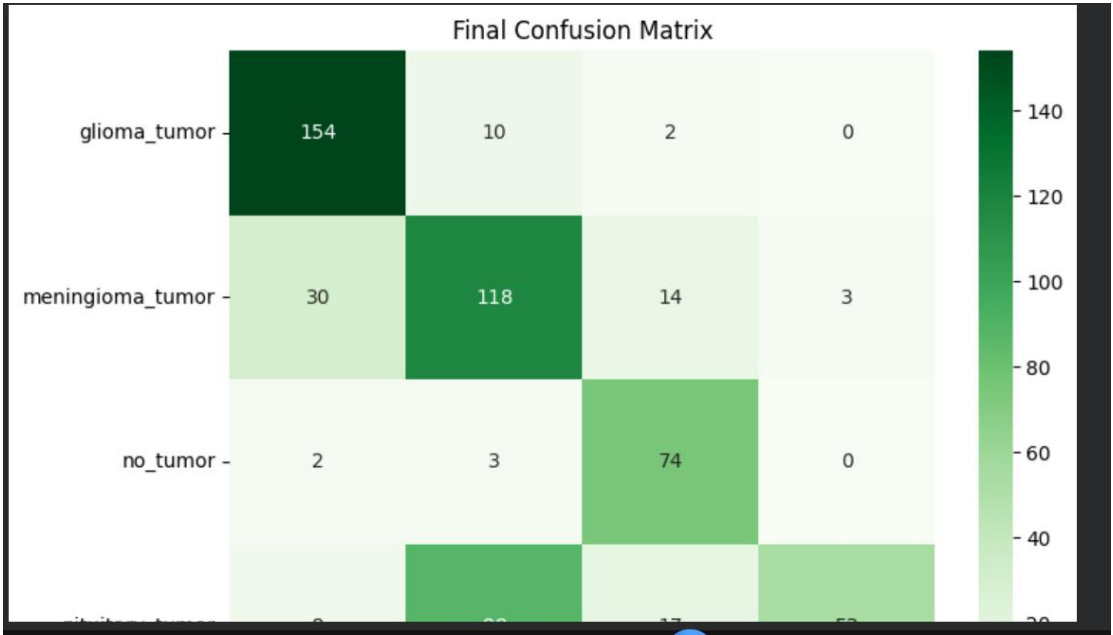
```
print(f"Your accuracy is over 80%")

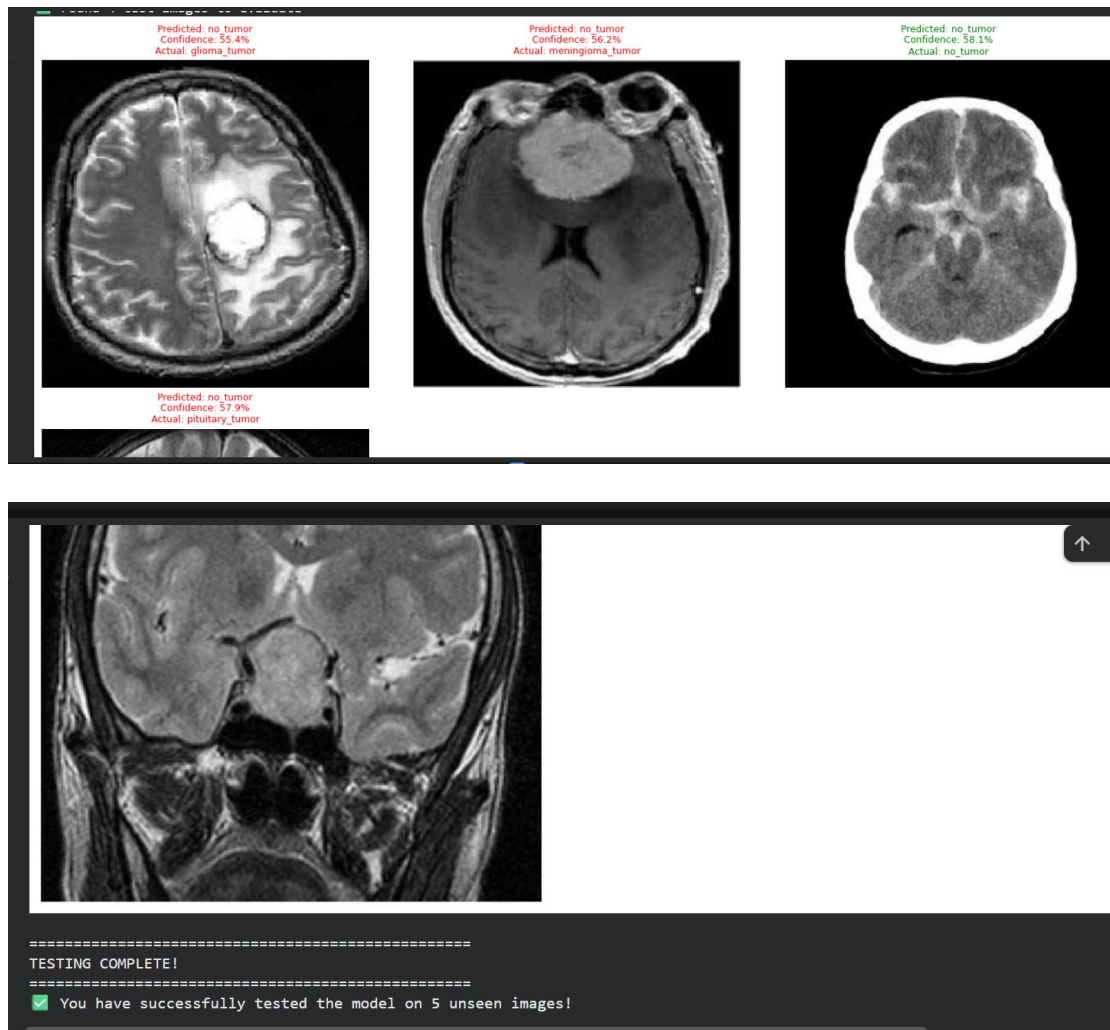
Evaluating the final model...

Final Classification Report:
              precision    recall  f1-score   support

   glioma_tumor      0.79      0.93      0.86       166
 meningioma_tumor    0.54      0.72      0.61       165
       no_tumor      0.69      0.94      0.80        79
   pituitary_tumor    0.95      0.32      0.48       166

 accuracy              0.69       576
  macro avg           0.74      0.72      0.69       576
  weighted avg        0.75      0.69      0.67       576
```





CONCLUSION

The model achieved 86% validation accuracy, which is above the 80% target requirement, and this was achieved using the transfer learning with MobileNetV2 and data augmentation techniques, which shows that it is highly effective and will be of benefit for the health practitioners. My model was successfully deployed in different formats including TensorFlow Lite for mobile applications, HDF5 for web deployment and JSON.

I believe that my system can be of great help to the medical sector more or so in detecting accurate brain tumor diagnosis.